

Assessment Method	Quiz / Assignment / ESE
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Suggested Reading:

1. "Research Methods in Artificial Intelligence" by Nik Bear Brown
2. Bryman, A., & Bell, E. (2015). Business research methods (4th ed.). Oxford University Press

Course number	MB - 542 (GAI)
Course Credit	L-T-P-C: 3-1-0-4
Course Title	Data Visualisation for Managers
Learning Mode	Online
Learning Objectives	☐ To understand the fundamentals of data visualization. ☐ To know the working principles of various information visualization depth tools. ☐ To acquire knowledge about the issues in data representation. ☐ · To visualize the Data using tools Tableau
Course Description	This course introduces the basic concepts of data analytics for understanding the importance of collecting, validating, managing and presenting data. The course helps to lay strong foundations on the tools and techniques for data management and visualization and aims to integrate them with those from allied management disciplines.
Course Outline	Context of data visualization – Definition, Methodology, Visualization design objectives. Key Factors – Purpose, visualization function and tone, visualization design options – Data representation, Data Presentation, Seven stages of data visualization, widgets, data visualization tools. Mapping - Time Series - Connections and Correlations - Scatterplot Maps - Trees, Hierarchies, and Recursion - Networks and Graphs. Mapping - Time series - Connections and correlations – Indicator-Area chart-Pivot table Scatter charts, Scatter maps - Tree maps, Space filling and non-space filling methods Hierarchies and Recursion - Networks and Graphs-Displaying Arbitrary Graphs-node link graph-Matrix representation for graphs- Info graphics. Acquiring data, - Where to Find Data, Tools for Acquiring Data from the Internet, Locating Files for Use with Processing, Loading Text Data, Dealing with Files and Folders, Listing Files in a Folder, Asynchronous Image Downloads, Web Techniques, Parsing data - Levels of Effort, Tools for Gathering Clues, Text Markup Languages, Regular Expressions, Grammars and BNF Notation, Compressed Data, Vectors and Geometry, Binary Data Formats, Advanced Detective Work. Drawing with data – Scales – Axes – Updates, Transition and Motion – Interactivity – Layouts. Port scan visualization - Vulnerability assessment and exploitation - Firewall log visualization - Intrusion detection log visualization -Attacking and defending visualization systems – Creating secured visualization system.